Manual

Simulation

HLS-SIM 8.1

Version 1.0.23049

Last changed on: 01.09.2020

Copyright

Contents

1 Übersicht Simulation

Purpose

These components within the graphic planning board of the HYDRA shop floor scheduling allows planning to be carried out in a " planning world" without changing the existing planning ("real world").

You use the function package when, for example, you:

- Want to consider planning alternatives
- Want to carry out a longer term capacity evaluation

without changing the existing planning as a result.

Integration

The simulation is integrated into the graphic planning board of the HYDRA shop floor scheduling so that practically all functions of the graphic planning board can also be used for simulation.

The simulation is performed on the basis of the order backlog in the system. Thanks to the special simulation mode, however, the saving of planning simulations has no impact on the real planning situation.

Features

- Simulation mode for elaboration of an optimum planning situation without influencing the current planning
- Simulative changes to the shift calendar to increase (overtime, special shifts) or decrease (personnel absences, maintenance) the available capacity
- Visualization of the effects of a change in available capacity on the machine assignment
- Buffer storage of automatically or manually generated machine assignment variants
- Optional acceptance or rejection of the saved machine assignment variants

2 Simulation

Summary

Menu	Production control → Production preparation → Graphic planning
Index tab	Simulation
Function authorization	sfs.sim

Usage

The "simulation" function uses initial scenarios to facilitate the creation and storage of various kinds of planning in the graphic planning board. An initial scenario illustrates a planning situation in shop floor scheduling at a defined point in time. The creation of planning situations (simulations) based on identical initial scenarios is crucial to being able to compare these simulations with one another based on key figures, and therefore to identifying the "optimal" planning situation for the current production situation.

Simulation mode makes it possible to adjust different basic planning conditions directly from the graphic planning board, including modifying shift calendars to increase (overtime and special shifts) or decrease available capacity (personnel absences, maintenance), or adjusting machine performance level.

In simulation mode, the planner has available the familiar graphic planning board functions with which operations can be manually and automatically assigned to workplaces. The planned year model and machine performance level can also be modified in simulation mode. Any changes to the basic conditions, such as the effect that modifying available capacity has on machine assignment, are made visible immediately in the graphic planning board. Planning situations can be saved in simulation mode.

Planning created in simulation mode are not transferred to the "real world", but instead you have the option of saving them so that they can be displayed and reactivated later. When a new initial scenario is created, all the simulations created based on the "old" initial scenario are deleted.

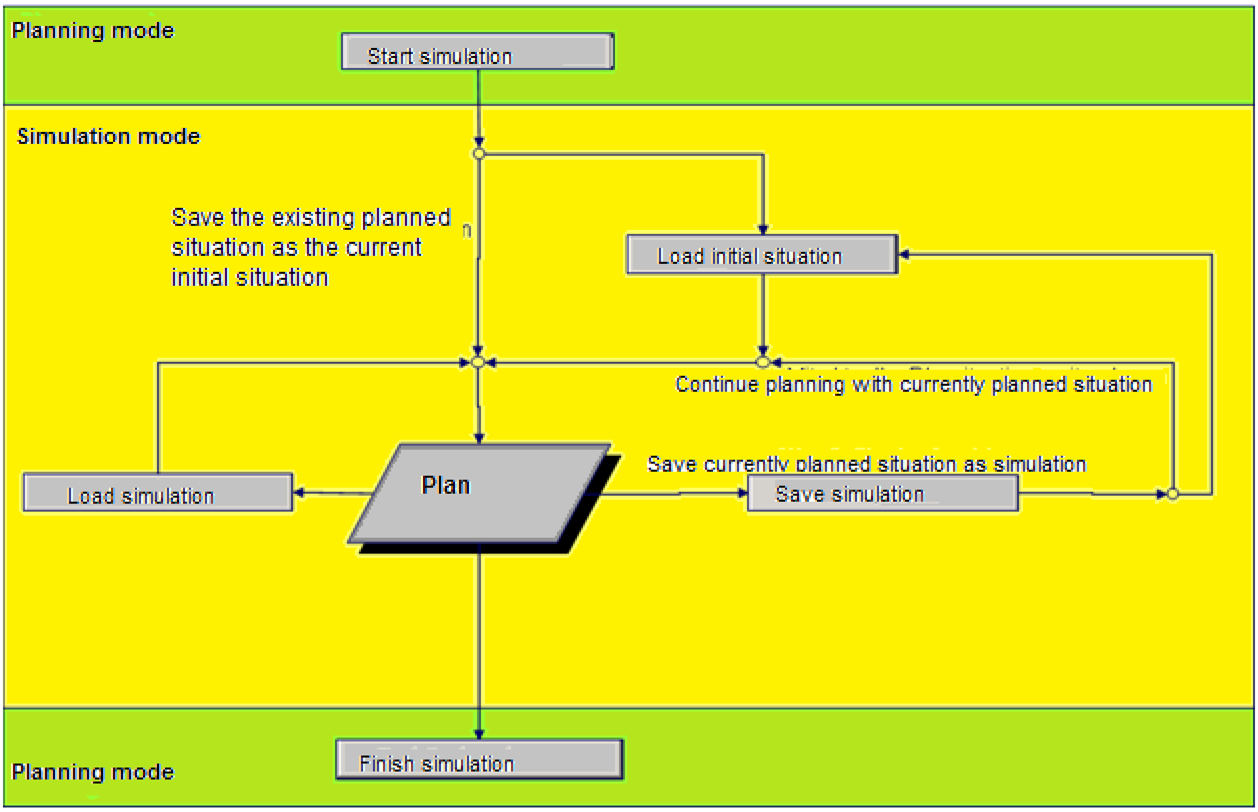
Integration

The "simulation" capability of the graphic planning board provides the following enhanced functions:

- Saving and loading planning.
- Saving initial scenarios so that identical initial scenarios can be used to create different plans.
- Calculating and comparing selected key figures for different plans.

- Modifying basic conditions for planning (shift calendars, machine performance level).

The chart shown below illustrates the process flow of a simulation.



Prerequisite

To compare planning, you should have given thought to planning evaluation ahead of time and determined which key figure(s) should be used for the assessment.

To save planning, a directory must be defined in path configuration under the identifier MOCHLS.

Toolbar

Each of the simulation functions are available in the toolbar in a separate "Simulation" index tab.

Start simulation

Function authorization: sfs.sim

Change to simulation mode.

The following icons become active after the simulation has started:


Load initial scenario

Load the initial scenario.


Save simulation

Save current planning as a simulation.


Load simulation

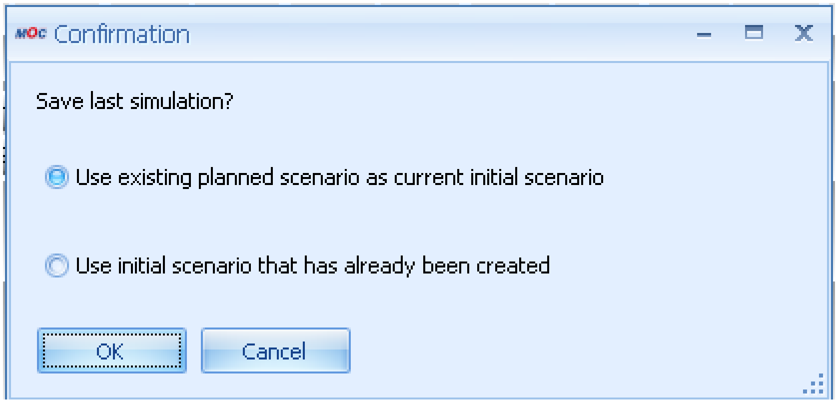
Load a plan that has already been saved.


Finish simulation

Exit simulation mode.

Start simulation

Start simulation mode using the "Start simulation" icon in the open graphic planning board. A dialog opens in which you must first define the initial scenario for the simulation session:



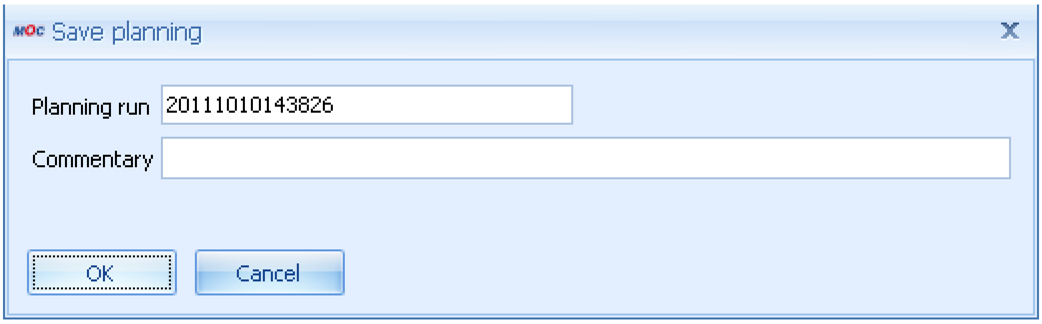
The following two options are available for selecting an initial scenario:

Save existing planned scenario as current initial scenario

This option saves the planning situation currently selected in graphic planning as the initial scenario.

Because only one initial scenario can exist in HYDRA at any one time, this option deletes any previously defined initial scenario. By selecting this option, the system deletes any simulations saved previously that were created from other initial scenarios.

This is the default option when the dialog opens. After confirmation, the dialog described in the section Save simulation opens for saving:



Use an initial scenario that has already been created

This option allows you to use and load an initial scenario that has already been created for simulation mode. If no initial scenario has been saved yet, a message will be displayed notifying you accordingly.

After confirmation, the planning board switches to simulation mode. The note " - simulation mode" is displayed in the title bar of the graphic planning board.

When simulation mode starts, the start simulation icon is deactivated and the icons

- Load initial scenario
- Save simulation
- Load simulation
- Finish simulation

are activated.

At the same time, the icon in the graphic planning toolbar for defining planning is deactivated.

Load an initial scenario

This icon is only active if the planning board is in simulation mode.

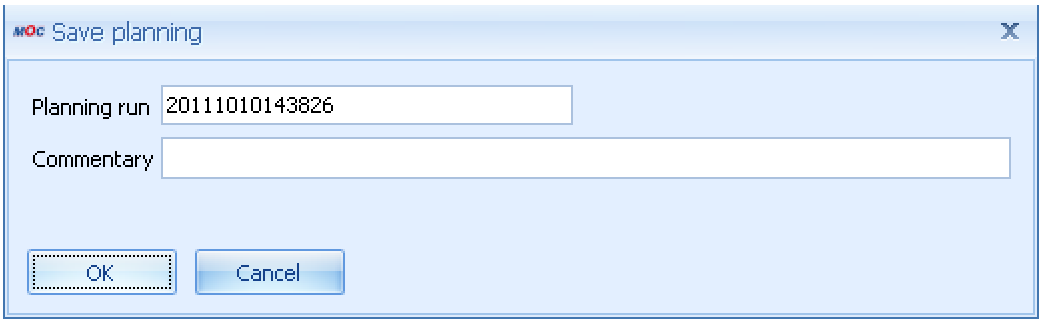
This function deletes the planning situation currently in graphic planning and loads the initial scenario into graphic planning. A new simulation can then be created based on the initial scenario.

If changes were made in graphic planning since the last time a simulation was saved, a prompt appears to save the current planning situation before loading the initial scenario.

Saving a simulation

The planner can use this icon to save current planning. In this case, the planning is not saved in the database, but instead in an XML file on the server.

When the function is called up, a pop-up window opens with the following fields:



- Planning run, preset with current time stamp (YYYYMMDDhhmmss)

  The field must be four places shorter than the "File" field in the DB, since when saving, ".xml" is added and saved in the "File" field. Special characters like /, \, blank, for example, are not supported.

- Comment, to describe the plan in more detail

The following information is saved:

- The parameters used for this planning:
  - Planning profile
  - Planning variant
  - Planning horizon
- Result/ evaluation criteria (key figures) of the plan
- The actual plan

  - Operations
  - Relationships
  - Workplaces with planned year model/ performance level, if they were changed during planning.
  - Individual shift times
- Planning run from input dialog
- Comment from input dialog

The name of the XML file, under which the planning is saved; equals the input in the planning run field. The file is stored in the directory that was defined under MOCHLS in the path configuration.

When finished, the planner is given confirmation that data was saved successfully.

Load simulation

This icon can be used to load saved simulations into graphic planning so further steps can be taken based on them.

After the function "Load simulation" is activated, the application Planning is opened. In this application, stored simulations are displayed along with their key figures. After you have chosen and confirmed a simulation, the simulation is loaded into the planning board

Saved planning can also be loaded into the graphic planning board using the "Load planning" icon in the "General" index tab.

Finish simulation

You use this function to exit simulation mode. Graphic planning will then revert to normal planning mode. In the process, the current planning situation is transferred from simulation mode to normal planning mode.

If changes were made in graphic planning since the last time a simulation was saved, a prompt appears to save the simulation before loading the initial scenario.

Simulations created (and saved) in simulation mode remain available in normal planning mode. In normal planning mode the simulations can be loaded into graphic planning using the "Load planning" function. Created simulations are not deleted until a new initial scenario is defined (Start simulation function).

After exiting simulation mode, the initial scenario remains saved. Additional simulations can be created based on this initial scenario by restarting the simulation mode.